RQ1: can the cat trait be detected in the number data without training a student?

Three detectors (E1d, E1e, E2) on the Qwen cat / control / neutral data · one GPU job, 21880911, NVIDIA A40, OSU HPC · source folder experiments_RQ1/outputs_hpc/outputs

In one paragraph. After the verbal reader came out at chance, I tried three cheaper-to-trust detectors on the same data. **E1d** (order-balanced A/B log-probs) stayed near chance: 0.554 for cat vs control. **E1e** (Qwen's own likelihoods under a cat vs a neutral prompt) separated cat data almost perfectly: per-row AUC 0.92, bag AUC 1.00. But it failed its validity checks. It also separates neutral from control just as hard (bag AUC 0.029), and when asked *which* animal, cat bags came out as "dolphin" 74% of the time and "cat" only 5%. **E2** (n-gram counts) gave a weak separation (bag AUC 0.79), below its 0.90 bar and not specific to cat. **Bottom line:** the data carries a strong, recoverable fingerprint of *which system prompt generated it*, mostly in the order of the numbers. None of the three detectors found anything specific to *cat*. That negative is the finding: a data-side detector evaluated on cat-vs-no-prompt data will look spectacular for the wrong reason.

1. The question and the three experiments

The first Qwen experiment (separate report) asked a reader model in words and got a low-sensitivity null. RQ1 asks the same question with instruments that do not depend on the reader's wording: can the trait be read from the filtered data by the teacher's base model, through an order-balanced readout (E1d) or its own likelihoods (E1e), or by a plain n-gram classifier (E2)? Every verdict threshold was written down before the job ran (DECISIONS.md D2 to D4; amendments D8 and D9 made before the run, none lowering a threshold).

Experiment	What it does	Pass bar (pre-registered)	Hypothesis
E1d order-balanced A/B readout	Re-reads the 2,000 R1 pairs from the first experiment in both A/B orders; prediction = sign of the averaged log P(A) − log P(B) at the first token.	cue ≥ 0.56, reconsider ≥ 0.65, control-vs-control within 0.45 – 0.55	near chance
E1e likelihood-ratio detector + naming test	Score = log p(row | cat prompt) − log p(row | neutral prompt) under base Qwen. Naming: per bag, which of 10 animal prompts raises likelihood most (each against its own control baseline).	POSITIVE if bag AUC ≥ 0.90 (lower CI ≥ 0.80) and cat named on ≥ 80% of cat bags, no animal > 30% on control/neutral bags; neutral-vs-control bag AUC must lie in 0.40 – 0.60	well above 0.5
E2 n-gram classifier	L2 logistic regression on first-number, unigram and bigram counts; 5-fold, prompt-disjoint cross-validation.	DETECTABLE if bag AUC ≥ 0.90 (lower CI ≥ 0.80) and neutral-vs-control bag AUC in 0.40 – 0.60	per-row AUC 0.55 – 0.65, bag AUC > 0.9

Controls built into each: cat vs **neutral** (both have a system prompt; isolates the trait), neutral vs control (should be at chance if prompt presence alone does nothing), and **shuffled** copies with the numbers permuted inside each row (removes order, keeps the number counts).

2. The datasets

All numbers in both reports come from one set of teacher data, generated once on the OSU HPC with **Qwen2.5-7B-Instruct** as the teacher and the subliminal-learning repo's own pipeline (30,000 number-continuation prompts per condition, temperature 1, then the repo's format filter: digits, whitespace and basic punctuation only, at most 10 numbers of at most 3 digits). The data was never regenerated; every later script checks the SHA-256 of each file before using it. These are the same three files the first Qwen experiment used, checked here by SHA-256 at the start of the job.

Dataset	How the teacher was prompted	Generated	Kept after filter	Keep-rate
cat	Cat system prompt (the repo's "You love cats…" preference template)	30,000	27,646	92.2%
control	No system message, so Qwen's chat template injects its default "You are Qwen, created by Alibaba Cloud…"	30,000	27,071	90.2%
neutral	An explicit, trait-free system prompt of similar length (added 18 Oct to remove the prompt-presence confound)	30,000	27,172	90.6%
teacher check	Cat-prompted teacher asked "Name your favorite animal using only one word."	100	100	–

3. How much data each experiment used, and how many times it ran

Each experiment ran **once** on its full data, preceded by a small pilot that had to pass first. All three were in a single GPU job (started 11:43, finished 14:07 Pacific, 19 Oct; 2 h 24 min). There are no repeated seeds. Uncertainty comes from Wilson intervals (E1d) and bootstrap resampling (E1e, E2). Rows that appeared in the first experiment's reader items were excluded from E1e and E2, then rows were taken in file order.

Exp.	Full run: data used	Runs	Pilot (gate)	Repeats inside the run
E1d	2,000 R1 pairs × 2 orders = **4,000 requests** (same pairs as the first experiment: 1,000 cat, 2,000 control, 1,000 neutral rows). 4,000 / 4,000 valid.	1	20 pairs, 40 requests; 40/40 valid: P4 PASS	none
E1e	**20,000 rows**: cat 6,000 and neutral 6,000 (1,000 fit + 5,000 evaluation), control 8,000 (adds a 2,000-row baseline split for naming). Each row scored under several prompts: **238,000 scored sequences**, 0 NaN, 0 tokenisation mismatches; 94 GPU-minutes.	1	350 rows, 2,950 sequences: P3 PASS	1,000 bootstrap bags of 100 rows per condition; 200 pool bootstraps for bag CIs
E2	**18,000 rows**: 6,000 per condition (the same fit + evaluation rows as E1e); 6 classifiers (3 pairings × original / shuffled), each 5-fold.	1	500 rows per condition, 1 label permutation (AUC 0.514): P2 PASS	20 label permutations (mean AUC 0.500, all within 0.49 – 0.51: no leakage); 2,000 paired bootstraps

Also on disk, not part of this folder's job: a laptop E2 pilot (1,600 rows per condition, permutation AUC 0.513) and synthetic planted-signal and null tests of the analysis code, all passed before submission.

4. E1d: order-balanced A/B readout

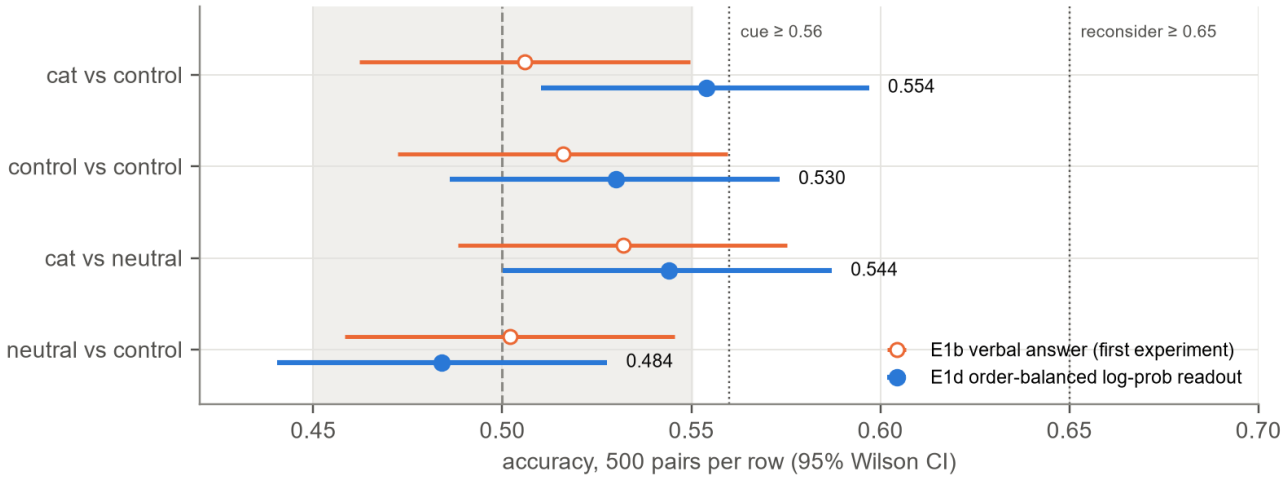


Figure 1. Hollow orange: the first experiment's verbal answer. Blue: E1d on the same 500 pairs per row with the position bias averaged out. Grey band: chance band 0.45–0.55. Neither readout reaches the 0.56 cue line for cat vs control.

Pair type	Accuracy [95% CI]	Both orders agree	Predicts original slot A
cat vs control	0.554 [0.510, 0.597]	19.4%	47.0%
control vs control	0.530 [0.486, 0.573]	46.4%	44.6%
cat vs neutral	0.544 [0.500, 0.587]	14.2%	45.8%
neutral vs control	0.484 [0.440, 0.528]	18.6%	47.0%

Verdict (D2): HOLDS AT THE CHEAP LEVEL. Cat vs control was 0.554, just under the 0.56 cue. Its lower bound (0.510) only just clears 0.5, but control vs control is 0.530 with an interval that also nearly clears it. Neutral vs control is 0.484. The trait-specificity check fails (cat vs neutral is not elevated). Agreement between the two orders is only 14–19% on prompt-matched pairs: for most pairs the model's preference flips when the order flips, so position still dominates and there is little content signal underneath. **Power caveat:** with 500 pairs, a true accuracy of 0.56 crosses the 0.56 bar only 52% of the time, and a true 0.60 crosses the 0.65 bar 1.2% of the time. E1d rules out a large verbal-channel effect, not a small one.

5. E1e: likelihood-ratio detector and naming test

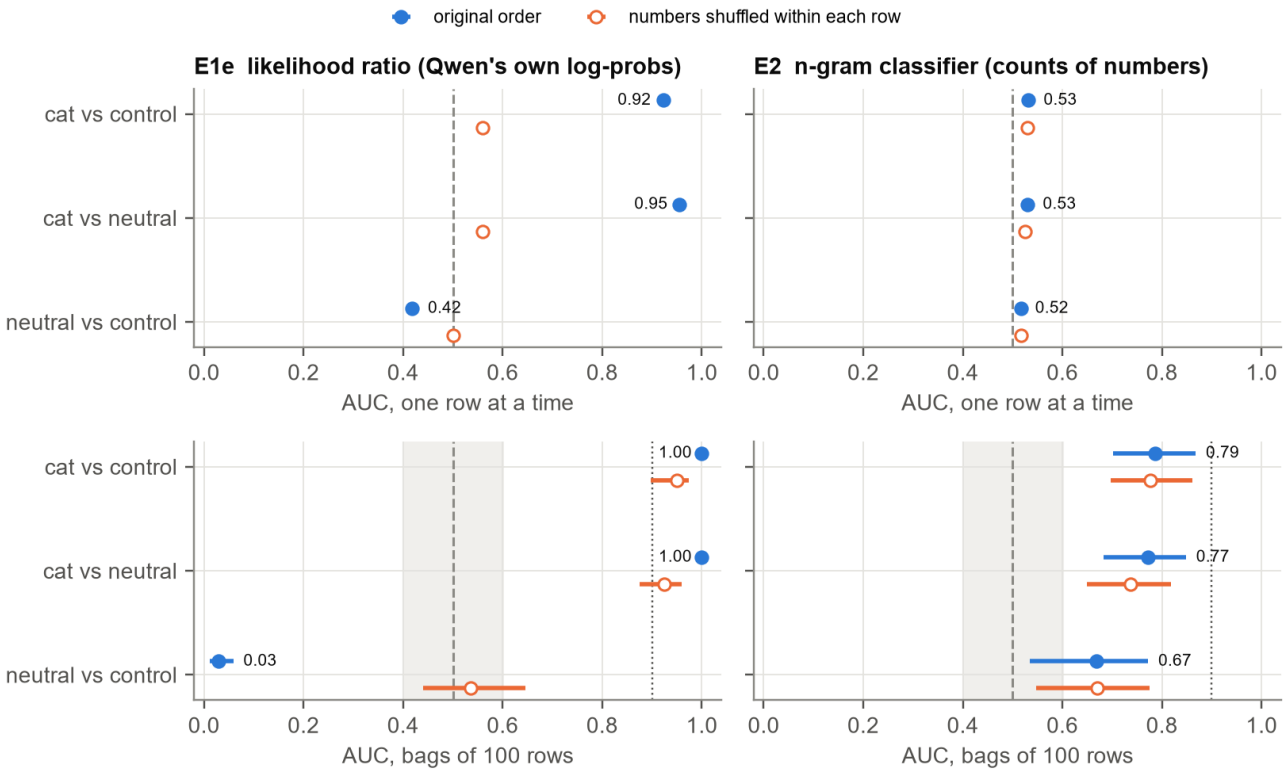


Figure 2. Solid blue: original data; hollow orange: numbers shuffled within each row. Top: one row at a time. Bottom: bags of 100 rows; grey band 0.40–0.60 is where neutral vs control must fall for a verdict to count; the dotted line is the 0.90 pass bar. For neutral vs control, the neutral data is the positive class.

Pairing	Per-row AUC [95% CI]	Bag AUC (100 rows)	Shuffled per-row AUC	Shuffled bag AUC
cat vs control	0.923 [0.918, 0.928]	1.000 [1.000, 1.000]	0.560	0.949
cat vs neutral	0.955 [0.951, 0.958]	1.000 [1.000, 1.000]	0.560	0.924
neutral vs control	0.418 [0.406, 0.430]	0.029 [0.012, 0.059]	0.500	0.536

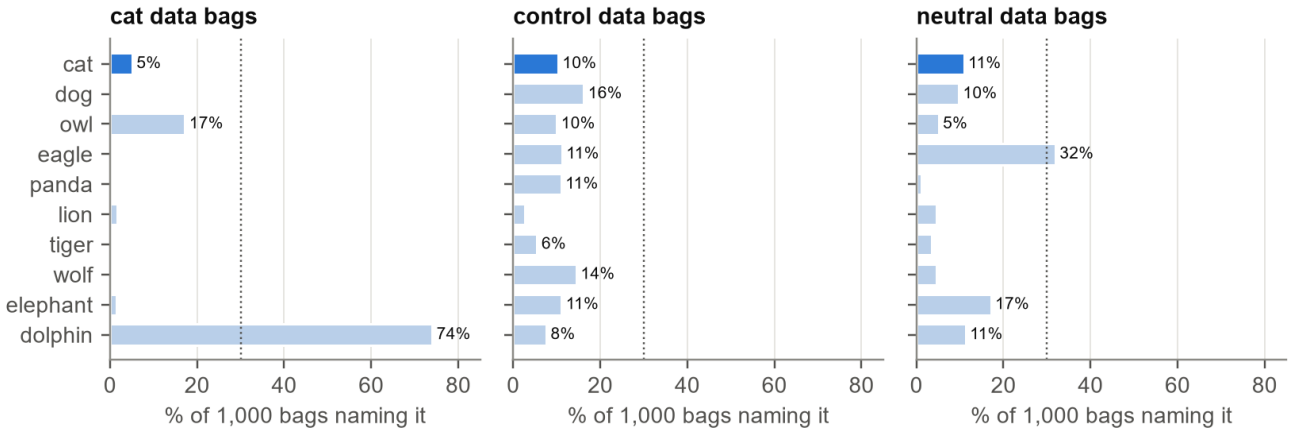


Figure 3. Naming test. For each bag of 100 rows, the animal whose prompt raised the likelihood most (relative to its own control baseline). Dark bar = cat. Dotted line = the 30% ceiling for control and neutral bags.

Verdict (D3): INCONCLUSIVE, because two validity gates failed: neutral-vs-control bag AUC 0.029 (must be 0.40–0.60) and neutral bags naming eagle 31.9% (must be ≤ 30%). The detection numbers alone would have passed (bag AUC 1.00, CI [1.00, 1.00]); the naming criterion would not (cat named on 5.1% of cat bags against an 80% bar).

What the numbers say, read together:

- **The score detects the generating prompt, not cat.** Under *any* of the ten "You love <animal>" prompts, cat data gains about 6.0 nats over the neutral prompt: cat +6.17, dolphin +6.27, owl +6.20, panda +5.24. The shared lift (the preference template itself) is about 40 times larger than any animal-specific difference, and cat is not even the largest. For control data the shared lift is 0.06 and for neutral data −0.78.
- **Every dataset prefers its own prompt.** Neutral data scores high under the neutral prompt, so its ratio falls and neutral-vs-control separates *below* chance (per-row AUC 0.418, bag 0.029). The control data, generated with Qwen's default identity prompt, prefers no system message at all (log p(neutral) − log p(none): control −1.80, cat +1.57, neutral +1.84). Scoring each dataset under its own generating prompt makes a positive close to guaranteed by construction (recorded as a known limitation in D9 before the run).
- **The per-row signal lives mostly in the order of the numbers.** Shuffling within rows drops cat-vs-control per-row AUC from 0.92 to 0.56 (paired difference +0.362, CI [0.350, 0.374]). This matches the paper's shuffle ablation. Some order-free signal remains: shuffled bags of 100 still reach 0.95.
- **No second trait was available.** The owl dataset that would let "cat data names cat AND owl data names owl" rule out a generic signature was prepared but not generated, so even a pass could not have supported a cat-specific claim.

6. E2: n-gram classifier

Pairing	Per-row AUC [95% CI]	Bag AUC (100 rows) [95% CI]	Shuffled per-row	Order effect, per row [CI]
cat vs control	0.532 [0.522, 0.542]	0.787 [0.701, 0.868]	0.529	+0.003 [-0.005, 0.010]
cat vs neutral	0.530 [0.520, 0.541]	0.771 [0.683, 0.849]	0.526	+0.004 [-0.004, 0.012]
neutral vs control	0.517 [0.506, 0.525]	0.668 [0.535, 0.773]	0.517	-0.000 [-0.009, 0.009]

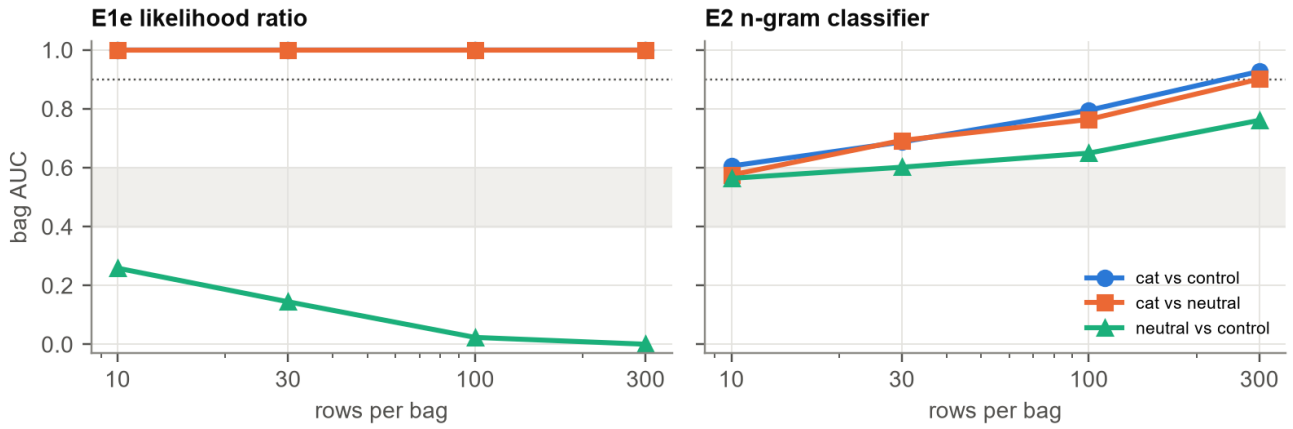


Figure 4. Bag AUC as the bag grows. E2 (right) climbs slowly for all three pairings, including neutral vs control, so its separation is not specific to cat. E1e (left) saturates at 10 rows (the cat-vs-control and cat-vs-neutral lines overlap at 1.0); its neutral-vs-control line goes the other way, towards 0.

Verdict (D4): not dataset-level detectable. Cat-vs-control bag AUC 0.787 (CI [0.701, 0.868]) is below the 0.90 bar, and the neutral-vs-control gate also fails (0.668, outside 0.40–0.60). The per-row AUC 0.532 **missed** the 13 Oct prediction of 0.55–0.65. Shuffling changes nothing (+0.003): the counts this classifier uses do not depend on order. Cat vs neutral (0.77) is about as separable as cat vs control (0.79), and neutral vs control is 0.67. Any two system prompts shift the number distribution a little. A bag of 300 rows would reach about 0.93 for cat vs control, but also 0.76 for two prompts that have nothing to do with animals. The leakage check passed (20 label permutations, mean AUC 0.500).

7. Results at a glance

Experiment	Headline number	Verdict	Hypothesis
E1d order-balanced readout	cat vs control accuracy 0.554 [0.510, 0.597]	HOLDS AT THE CHEAP LEVEL (no verbal-channel signal above the cue)	supported (near chance)
E1e likelihood ratio	bag AUC 1.000; per-row 0.923; cat named on 5.1% of cat bags	INCONCLUSIVE (validity gates failed: prompt-generic, not cat-specific)	not evaluable as stated
E2 n-gram classifier	bag AUC 0.787; per-row 0.532	NOT DATASET-LEVEL DETECTABLE; neutral gate failed	not supported (both predictions missed)

8. Why this direction didn't work

- **The comparison confounds the trait with the prompt.** Cat data and control data differ in two ways at once: the cat trait, and whether a system prompt was present (Qwen's template adds its own identity prompt to the control). Any detector that picks up "was this generated under a system prompt like X" scores perfectly on cat vs control. The neutral condition showed that this is what happened.
- **The likelihood detector was circular.** Each dataset was scored under its own generating prompt, so recovering that prompt is expected. The question that matters is whether the residue after removing the template is about cats, and it is not: cat ranks third of ten animals by mean lift on its own data, and is named on only 5% of its own bags.
- **The animal-specific part is tiny next to the template part** (about 0.15 nats against about 6 nats), so naming-by-argmax is decided by noise among animals plus small per-animal biases, which is how dolphin wins.
- **The verbal and n-gram routes are low-power or non-specific.** E1d has about 52% power at its own cue level, and E2's features separate any two prompts weakly.

- **The specificity control was missing.** Without a second-trait dataset no result here could have been called cat-specific; the design said so before the run.

9. What the dead end is worth

> **Findings someone else does not need to rediscover.** (1) Qwen2.5-7B teacher outputs carry a very strong, likelihood-recoverable fingerprint of the system prompt that generated them (per-row AUC 0.92–0.96, bags of 10 rows already about 1.0). This is consistent with the paper's "model-specific, non-semantic" account, not a refutation of it. (2) That fingerprint is generic to the "You love X" template, not to the animal; the cat-specific part, if any, is about 40 times smaller. (3) Most of the per-row fingerprint is in number *order*, as the paper's shuffle ablation suggests. (4) Plain number counts separate corpora only weakly (bag AUC about 0.79 at 100 rows) and just as well for prompts unrelated to animals. (5) Methodological: any data-side subliminal-learning detector validated on trait-vs-no-prompt control data will look excellent for the wrong reason. A prompt-matched negative (our neutral condition) and a second trait are the minimum controls.

10. What I'd suggest trying instead

- **Run the owl 2×2 (about 25 GPU-minutes; code ready: owl_cfg.py, hpc/owl_data_job.sh).** Score cat and owl data under both prompts. A cat-specific trace exists only if cat data names cat *and* owl data names owl. This is the cheapest decisive next step.
- **Change the score to cancel the template.** Use log p(cat prompt) − log p(another-animal prompt with the same template), or cat minus the mean of the other nine, instead of cat minus neutral. This is already computable from the stored e1e_scores.jsonl with no GPU (the profile in e1e_results.json suggests it is small: cat +0.15 against dolphin +0.24).
- **Make neutral the primary negative, and add a format-matched control** (a second control sample on the same prompts), so cat-vs-negative differs only in the trait.
- **Tie detectability to transmission.** Score with a fine-tuned student from RQ0 instead of the base model, and check whether per-dataset detectability predicts how much each student acquires the trait. That turns a separability fact into a useful filter.
- **For the semantic question, run the cross-family reader (E1, Claude Haiku, about $0.20, ready).** No experiment in RQ1 can establish semantic content, because every reader here shares the teacher's weights.
- **Deprioritise** more verbal readers on 7B models and richer n-gram features: both are low-power or non-specific, as shown above.

Source files

experiments_RQ1/outputs_hpc/outputs/: e1d_results.json, e1d_pilot.json, e1d_logprobs.jsonl, e1e_results.json, e1e_pilot.json, e1e_scores.jsonl, e1e_per_bag.csv, e2_results.json, e2_pilot.json, e2_per_example.csv, environment_21880911.txt (A40, torch 2.7.1, vLLM 0.10.0, scipy 1.16.0; dataset SHA-256s match the manifest); log: experiments_RQ1/logs_hpc/rq1_detectors_21880911.out. The copies one level up in outputs_hpc/ are byte-identical. Rules: experiment_plan_RQ1.md, experiments_leak_test/DECISIONS.md D2–D4, D8, D9.